

Infrastructure and accessibility implications of implementing x-minute city policies in low-density contexts

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ABSTRACT

Addressing priorities such as health, liveability, and climate resilience, many global governments are exploring x-minute city and x-minute neighbourhood policies. Addressing a lack of operational models for x-minute cities, we developed a model to assess accessibility and utilisation implications of their implementation, testing the model by applying it to the low-density city of Melbourne, Australia, where 20-minute neighbourhood (20MN) policy focuses on walkable access to daily destinations within 10 minutes (20 minute round trip), and applying the model to people living within 10-minute walking distance ‘catchments’ of mixed-use activity centres. We curated a list of 14 destination types (such as supermarkets and primary schools), and developed a method to model notional placement of additional destinations to place at least 80 % of people in each catchment within a 10-minute walk of each destination type. We extended accessibility considerations to cycling, based on a 10-minute one-way ride. Our results show improvements in accessibility across the city as a whole, particularly in inner urban areas; but with significant improvements in outer urban areas, characterised by low housing density, when cycling is promoted. Our utilisation analysis demonstrates feasibility challenges when implementing x-minute city policies in less densely populated locations. Our conclusions underscore the importance of supporting x-minute city policies with safe cycling infrastructure and careful urban densification strategies. Our findings are relevant to many cities seeking to implement x-minute city or neighbourhood policies worldwide, especially cities facing challenges of low density.

1. Introduction

1.1. Neighbourhood planning and sustainability

Population growth, an aging population, congestion, and housing affordability are all significant challenges to sustainability faced by cities globally. Strategic urban planning is essential to confront these issues, prioritising the development of healthy, liveable, and climate-resilient cities (Chu et al., 2017). City officials worldwide are addressing these challenges by formulating urban planning strategies to increase proximity to local destinations within neighbourhoods (Logan et al., 2022; Scottish Government, 2023; Sepehri & Sharifi, 2025; State Government of Victoria, 2019). Reshaping cities to support local living

has co-benefits. It will improve public health by facilitating physical activity, social interaction, and economic vitality, which align with broader objectives of sustainable urban development (De Roo & Miller, 2019). Reducing reliance on private cars can mitigate traffic congestion, and is vital for improving air quality, reducing noise pollution and mitigating climate change (AlWaer & Cooper, 2023; Pozoukidou & Chatziyiannaki, 2021).

The recognition that compact, higher density urban development can address many complex sustainability challenges facing cities is not new (Pineo, 2022). In the past decade there has been a resurgence in interest in developing neighbourhoods that provide access to the amenities, services and employment that urban populations require within short distances easily reached by active transport modes such as walking and

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cycling (AlWaer & Cooper, 2023). By supporting local living, such ‘x-minute cities’ would encourage healthier and more active lifestyles, reducing car dependency and associated emissions (Logan et al., 2022). This is achieved by providing easy access to daily living destinations including social and commercial infrastructure, such as grocery shops, health care and recreational opportunities, using active transport, as well as access to city-wide services and employment via high-quality public transport (AlWaer & Cooper, 2023; Mackness et al., 2022; McNeil, 2011). Population density is crucial to achieving compact city layouts with local environments capable of supporting diverse local destinations (McNeil, 2011). Without enough people within walkable catchments, delivering local destinations and high quality public transport becomes less economically viable (Kim & Woo, 2022; Sohn et al., 2012).

With lockdowns and movement restrictions, the COVID-19 pandemic highlighted the importance of neighbourhoods as the building block of healthy and sustainable cities, and the need for improved urban and transport planning (Nieuwenhuijsen et al., 2022), including equitable access to local destinations and amenities. It also led to temporary and permanent reassignment of road space (Calafiore et al., 2022; Frank et al., 2021). The pandemic prompted a global commitment from city mayors to ‘build back better’ through the creation of 15-minute cities (C40 Cities Climate Leadership Group Inc, 2024), a sustainable urban development concept popularised by Moreno et al. (2021), aimed at ensuring that all residents have 15-minute access to services required to live, learn and thrive, including employment (Moreno et al., 2021).

1.2. Global adoption of x-minute city concepts

Rethinking city planning to create x-minute cities appears to have captured the imagination of researchers and policy-makers globally. X-minute cities have the potential to address pressing global concerns including greenhouse gas emissions (Allam et al., 2022) local liveability, health and wellbeing (Schendl et al., 2025; Stevenson et al., 2025) and motor vehicle dependency (Traore et al., 2025). Haque et al. (2025) have reviewed x-minute neighbourhood policies across a range of cities in Europe, Canada, the USA, Australia, New Zealand and Singapore, while the Chinese government has also promoted 15-minute walkable neighbourhood policies (Weng et al., 2019). Nevertheless, to date, x-minute city research efforts appear to be underrepresented in some regions including the Middle East and Africa (Sepehri & Sharifi, 2025).

Despite some public scepticism and conspiracy theories circulating about government motives behind x-minute cities (Marquet et al., 2024), many cities across the globe appear to have adopted the concept. Logan et al. (2022) found that 14 cities worldwide that have adopted x-minute city policies, ranging from European high-density cities such as Copenhagen, Paris and Milan, to lower-density cities across North America and Oceania such as Christchurch (NZ), Portland (Oregon, USA), Detroit (Michigan, USA) and Ottawa (Canada). Similarly, Lu and Diab (2023) compared x-minute city policy documents across a range of US, Canadian and Australian cities, many of which included low-density environments, while Pozoukidou and Chatziyiannaki (2021) compared policies in the high-density city of Paris with those in the low-density cities of Portland (Oregon, USA) and Melbourne (Australia).

Nevertheless, despite some policy-uptake, the ‘devil is in the detail’ about how to achieve equitable x-minute cities. Some have argued that successful x-minute cities require increased population and services density (Allam et al., 2023; Jafari et al., 2023). This is because the success of x-minute cities depends on the presence and viability of local destinations; and the viability depends on having sufficient residents to support local services and businesses.

Hence, recent studies have shifted from ‘what’ and ‘why’ x-minute cities are important, to ‘how’ they can be achieved. Studies have explored levels of population density required to ensure all residents have access to amenities (Jafari et al., 2023) and public transport (Mouratidis, 2026) within x-minutes; the contribution of road networks

in opening up 15-minute cities (Fan et al., 2025); the impact of types of employment and transport performance (Li et al., 2025); the importance of integrating both transit accessibility and connectivity in achieving 15-minute cities (Liu et al., 2025); the role of distribution parcel lockers in reducing the environments impacts of freight transport in an x-minute city (Orhan & Wallace, 2025); the role that cycling (Both et al., 2022) and bike sharing (Petert et al., 2024) could play in achieving x-minute cities; the need to consider mobility patterns (Zhang et al., 2025); and whether the availability of amenities alone is sufficient to change consumer behaviour (Maciejewska et al., 2025). To date, research does not appear to specifically evaluate specific government efforts to achieve x-minute cities. This paper seeks to address this gap by developing a methodology for natural experiments assessing x-minute policies (Craig et al., 2012).

1.3. Scope and research questions

As urban and transport planning policies in low-density cities endeavour to facilitate the x-minute city concept, evidence gaps remain concerning the potential for operationalising these policies and assessing their impacts. From an operationalisation perspective, should the policy be fully implemented, the gaps include the identification of additional destinations required, levels of accessibility that will be achieved, and the potential utilisation of these destinations by residents within a walkable catchment. From a theoretical perspective, the questions arise whether x-minute cities are possible in a low-density context given low utilisation of destinations, and how the effects of low utilisation may be mitigated. Given the challenge of delivering x-minute neighbourhoods in established areas in low density cities, these issues lead to a need to investigate the potential contribution of cycling to their successful implementation (Both et al., 2022).

In this paper we propose and critically evaluate a method for modelling how destinations could be added around existing activity centres to achieve a desired accessibility target. We then propose a method to quantify the extent to which this could lead to under-utilisation of the new destinations if population density is not increased to ensure sufficient demand, and review strategies to ameliorate the consequences of underutilisation.

After describing the x-minute neighbourhood concept, this paper addresses the following research questions:

- Q1.** What additional destinations would be required to adapt the existing activity centres to implement an x-minute city policy?
- Q2.** How would the implementation of an x-minute city affect residents’ accessibility to destinations by foot and by bicycle?
- Q3.** In a low-density environment, what level of destination under-utilisation would be likely if an x-minute city were broadly implemented, without increases to population densities around existing activity centres?

These research questions are investigated using Melbourne, Australia as an illustration of a low-density city, with methods that are generalisable globally.

Section 2 provides a brief literature review on x-minute cities and neighbourhoods, including a brief overview of x-minute city policies, in the form of a 20-minute neighbourhood (‘20MN’) policy, in Melbourne. Section 3 outlines the research methods employed. Section 4 presents the findings, and Sections 5 and 6 include discussion, policy implications and consideration of the strengths and weaknesses of the analyses undertaken and areas for future research.

2. Background

2.1. X-minute cities: the concept

The x-minute city – or a city composed of x-minute neighbourhoods –

has been described as one where ‘residents are able to meet most of their needs within a short walk or bicycle ride from their homes’ (C40 Cities Climate Leadership Group Inc, 2020) or ‘...within a short walk from their homes’ (Logan et al., 2022).

From the ‘15-minute city’ conceived by Carlos Moreno in 2016 and piloted in Paris (Allam et al., 2023), the x-minute city concept has been adopted in cities across Europe, North America, Asia and Oceania (Levinson, 2019; Logan et al., 2022; Lu & Diab, 2023; Pozoukidou & Chatziyiannaki, 2021). Sepehri and Sharifi (2025) chart the growth in published articles dealing with the concept, from one article in 2019 to 89 meeting their search criteria in 2023, addressing domains spanning urban planning, sustainability, transport, accessibility, energy and health.

The x-minute city concept is linked with a broad range of environmental, economic, social and health benefits (Talen & Koschinsky, 2013). Neighbourhoods meeting the conceptual ideal are characterised by high residential densities, proximity to essential destinations, interconnected street networks, and a mix of diverse land-use, all of which support walking as a mode of transport (Hooper et al., 2020; Levinson, 2019; Smith et al., 2017).

As highlighted by Moreno et al. (2021), sufficient population density is critical for the commercial viability of shops and services, and thus for x-minute neighbourhoods. More compact, higher density and mixed-use developments are therefore essential for enabling the proximity of diverse services and amenities (Jafari et al., 2024; Morley & Pafka, 2023). Accessibility is typically defined by the distribution of destinations surrounding a specific location, the ease of reaching these destinations via different modes of transport, and the nature and quantity of activities present at these destinations (Geurs & van Wee, 2013; Handy, 2020). Provision of a mixed range of destinations, accessibility and population density are therefore three critical and interconnected dimensions that underpin the x-minute neighbourhood concept (AlWaer & Cooper, 2023; Thornton et al., 2022).

Addressing density, and turning to Melbourne as an example of a low-density city which has developed x-minute city policies, Boulange et al. (2017)’s cross-sectional analysis for Melbourne found that residing in areas with densities exceeding 20 dwellings per hectare was associated with higher rates of walking, cycling, and public transport use, and lower car usage, compared to areas with lower density. Other cross-sectional evidence from Melbourne also supports the positive association between higher population density and transport-related walking (Badland et al., 2017), and global studies show a higher rates of bicycle use in higher density areas (Pucher et al., 2011). Badland et al. (2017) found that as the distance to activity centres decreases, the odds of walking for transport increase and those with an activity centre within 1 km were over three times more likely to walk for transport. Their findings also indicated that with access to 3.4 additional types of local living destinations (e.g. grocery shops, parks), there was an 80 % greater likelihood of walking within an 800 m walkable network from home. Gunn et al. (2017) found that likelihood of walkability was enhanced most around activity centres with pedestrian-friendly design.

Policy requirements for provision of local access to a mixed range of service and amenity destinations to support x-minute neighbourhoods in Melbourne and other cities (Scottish Government, 2023; State Government of Victoria, 2019) typically include grocery shops, recreation, healthcare, education, dining out, and other essential amenities (Jafari et al., 2023; Olsen et al., 2022; Thornton et al., 2022; Weng et al., 2019). Implementation in some cities, for example the Nordhavn 5-minute city development in Copenhagen, may involve detailed curation of both commercial and social destinations to be included in the development (Grauslund Kristensen, 2025; Harris, 2025).

The transport modes forming part of the definition of the x-minute city concept vary between cities. Of the 14 city policies summarised by Logan et al. (2022), all refer to walking, while 9 also refer to cycling, and 5 to public transport. In this paper, we begin with walking, consistently with the Melbourne policies as discussed in Section 2.2, but we also

address the desirability of including cycling in the Melbourne policies as some other cities have done. This paper does not address the expansion of the x-minute concept to public transport.

2.2. X-minute cities: policy in Melbourne

Focusing on individual neighbourhoods (and not, at this stage, using the ‘x-minute city’ term introduced by Carlos Moreno), 20MNs were initially defined in Melbourne as those where people had access to destinations of daily living and employment centres within 20 minutes from home, whether by walking, cycling, or using public transport (State Government of Victoria, 2014). Subsequent definitions focused increasingly on walking, with the 2015 iteration of the plan emphasising destinations within 20 minutes, or an estimated distance of 1 to 1.5 km, and excluded working destinations (State Government of Victoria, 2017a). Later iterations of the concept in 2017 and 2019 refined 20MNs further, to be predominantly pedestrian-focused with an 800 m walkable catchment, implying destinations should be within a 10 minute walk, i. e. a 20 minute round trip (State Government of Victoria, 2019). The latest 20-minute neighbourhood strategic planning document, ‘Plan Melbourne,’ prioritises pedestrian accessibility. This includes proximity to essential daily living destinations within a walkable catchment such as schools, grocery shops, and healthcare services. While local employment was not included, the policy also emphasised access to employment and regional services facilitated through high-quality public transport (State Government of Victoria, 2017a). As part of this articulation of the 20MN concept, the Victorian Government identified an evidence-based target density of at least 25 dwellings per hectare as being necessary to achieve this level of amenity. The 20MN strategic planning indicated a departure from traditional policy and practice in Melbourne, where the emphasis previously had been on private car mobility, as evidenced by the extensive provision of road infrastructure, urban sprawl, and low population densities (Craven et al., 2014).

Activity centres in Melbourne serve as hubs offering essential services and infrastructure, such as retail shops, health services, and amenities, to meet the daily needs of nearby residents. These centres vary in size and offerings and could be strategically retrofitted with additional amenities to facilitate the creation of 20MNs, enabled by increasing local housing density to produce more diverse and affordable housing (State Government of Victoria, 2019). Achieving successful 20MNs requires a combination of amenities (destinations) and people to use them.

More recently, the Victorian government has begun to address housing density, by adopting statewide housing targets and an activity centres program. The statewide housing targets (State Government of Victoria, 2025c) provide each of Victoria’s 79 Local Government Areas (LGAs) with a draft numerical target for new homes to be built by 2051. Within Greater Melbourne, adopting the classification of LGAs shown in Fig. 5, these targets require increases in current housing numbers of 83 % to 122 % (Inner LGAs), 70 % to 106 % (Middle LGAs) and 33 % to 190 % (Outer LGAs). The activity centres program identifies 60 zones around train stations and tram stops in Melbourne where detailed plans are to be released for building more than 300,000 new homes (State Government of Victoria, 2025b). As of May 2025, the government had released plans for 10 zones (State Government of Victoria, 2025a), where maximum residential development heights are set at between 4 and 20 levels (depending on the zone), and additional commercial development is encouraged.

Thornton et al. (2022) assessed the 20MN concept in Melbourne, Australia, and Adelaide (the capital city of South Australia). They examined whether households had access to all five essential destination domains that characterise 20MNs, as derived from policies, evidence, and their own expertise. A household was considered as being in a 20MN if it was within pedestrian network distances of less than 1.5 km from all five destination domains. These domains included healthy food (fruit and vegetable market, supermarket), recreational destinations (gym), public open space (e.g. parks and gardens, sports fields), public transport

(bus, tram, train stop), and community resources (e.g. primary school, general practitioner (GP), pharmacy). The authors found that in Melbourne, 5.5 % of the population resided in 20MNs, and these were only where population and dwelling densities were the highest observed in the city. In a follow up study, [Contardo Ayala et al. \(2022\)](#) found that people living in 20MNs in Melbourne averaged 82.5 min of walking for transport per week in contrast to 41.2 min per week for those living in non-20MNs.

[Jafari et al. \(2023\)](#) explored the level of population density required to provide access to all of the destinations regarded as ‘hallmarks’ of a 20MN in Melbourne ([State Government of Victoria, 2019](#)) within a walkable 800 m catchment. Assuming 2.6 persons per dwelling, they found that at least 25 dwellings per hectare is needed to provide access to the destinations within 1.2 km and 35 dwellings per hectare to provide access within 1 km. They also found that the construction of low density 20MNs could lead to considerable underutilisation by residents within the 800 m walkable catchment, and hence unfeasible operational costs. Their findings indicated that by increasing neighbourhood densities from 15 to 30 dwellings per hectare, land requirements and building costs for destinations could be halved. Moreover, they concluded that to ensure access to all key destinations within 10 minutes (corresponding to 800 m in distance), it would be necessary to incorporate cycling infrastructure. This would enable residents to reach larger regional destinations with bigger population catchments, such as high schools, within 10 minutes.

Notably, existing peer-reviewed evidence on 20MNs in Melbourne, as well as other cities worldwide ([Gaglione et al., 2022](#); [Khavarian-Garmsir et al., 2023](#)), has primarily focused on characterising current levels of accessibility to 20MNs or related planning concepts, or on using optimisation models to define population density requirements for 20MNs in greenfield areas ([Jafari et al., 2023](#)). In our study, we extend this previous research and develop a methodology to assess an intervention where 20MNs are implemented anchored on existing activity centres in established areas of Melbourne. We explore the implications of implementing 20MNs in terms of destination requirements, including social and commercial infrastructure, levels of accessibility, and population requirements for new destinations.

Given that many activity centres are located in low density neighbourhoods, we also expand the scope of current 20MN policy to include cycling as a viable transport option, with the aim of exploring cycling’s potential contribution to realising 20MNs in Melbourne’s low density areas ([Both et al., 2022](#)). Cycling rates in Melbourne are currently low, with only around 2 % of trips made by bike and, as [Fig. 1](#) shows, the pattern is highly uneven across Melbourne, with inner-city neighbourhoods registering over 9 % while most outer suburbs sit below 1 % ([Jafari et al., 2025](#)). A growing body of scientific literature supports the concept that cycling can substantially expand x-minute access, if safe cycling infrastructure exists to support its uptake. For example, in Vancouver nearly the entire population has a grocery store within a 15-minute bike ride, whereas about one-fifth of residents lack one within a 15-minute walk ([Hosford et al., 2022](#)). Another study in north America found that in Victoria, Canada, the build-out of a low-stress cycling network nearly doubled the number of neighbourhoods where at least one active living, community, education, food, health, and shopping destination was accessible within 15 minutes by bike on low traffic stress routes between 2016 and 2023 ([Williams et al., 2024](#)). [Colaço and de Abreu e Silva \(2025\)](#) found that residents of 15-minute city neighbourhoods in Lisbon, Portugal, make more trips by walking or cycling to nearby supermarkets and parks. These findings show that well-planned cycling infrastructure and land-use proximity can create a great opportunity for accessing daily needs with active modes of transport. Recognising the importance of active modes for a 20MN city, the state and local governments in Victoria, Australia, have begun to prioritise cycling as a mode of transport to daily destinations. The Victorian Cycling Strategy 2018–28 commits the state government to work with councils to provide safe access to local destinations that underpin the 20MN

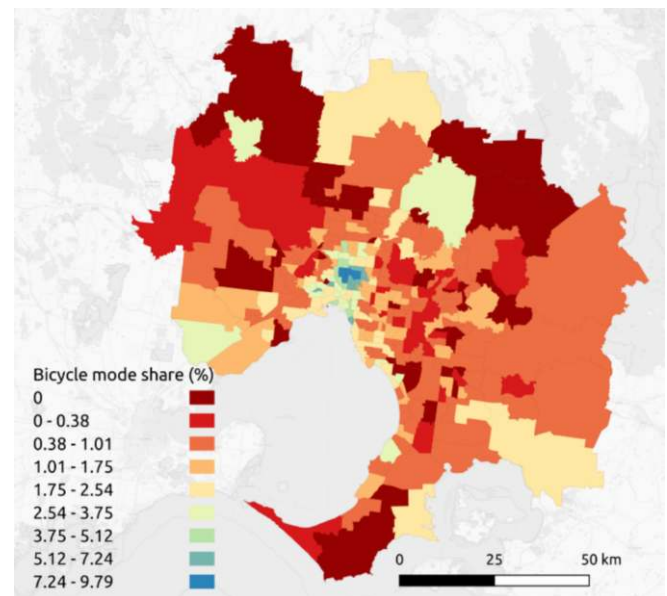


Fig. 1. Cycling mode share distribution across Greater Melbourne, basemap © OpenStreetMap contributors.
(Source: [Jafari et al. \(2025\)](#).)

concept, which is also reflected in the state government’s Strategic Cycling Corridors (SCC) program is setting out a metropolitan network of protected cycling routes to major destinations ([State Government of Victoria, 2017b](#)).

3. Methods

Our area of analysis was the Greater Melbourne Capital City Statistical Area delineated by the Australian Bureau of Statistics ([Australian Bureau of Statistics, 2016](#)), buffered by 10 km to allow for journeys that need to exit then re-enter the area.

We characterised 20MNs in Melbourne’s established areas based on current strategic planning policy ([State Government of Victoria, 2017a](#)), consultation with our Policy Advisory Group including state government and private sector members from the transport, planning and health sectors, and peer-reviewed evidence ([AlWaer & Cooper, 2023](#); [Calafiore et al., 2022](#); [Jafari et al., 2023](#); [Thornton et al., 2022](#)). The strategic planning for 20MNs in Melbourne in established areas is centered on establishing them around existing activity centres (ACs) of varying sizes, forming part of a citywide network of ACs ([Echelon Planning, 2022](#); [State Government of Victoria, 2019](#)).

Our workflow, illustrated in [Fig. 2](#), comprised 6 steps, which are respectively described in [Sections 3.1 to 3.6](#). The 6 steps are as follows.

1. *Activity centres and catchment areas.* We identified ACs and their population catchment areas: see [Section 3.1](#).
2. *Destinations and populations with access.* We determined destination standards for achieving 20MNs within the identified ACs and their catchment areas, specifying the proportion of the population that should have access to them by foot: see [Section 3.2](#).
3. *Baseline assessment.* We conducted a baseline assessment to determine the extent to which each AC met the target for each type of destination: see [Section 3.3](#).
4. *Creating 20MNs by adding destinations.* For those ACs falling short of the target, we developed a method to model the addition of missing destinations to meet the specified standards (the ‘intervention’): see [Section 3.4](#).

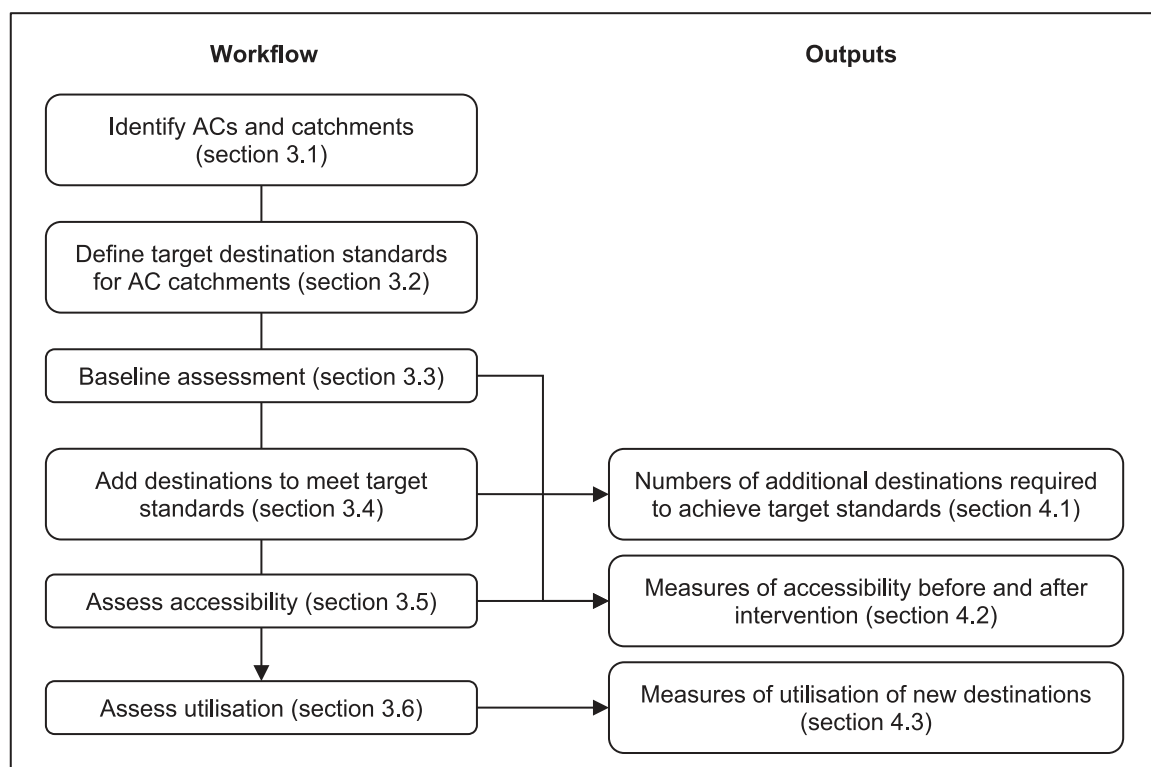


Fig. 2. Workflow diagram and key outputs.

5. *Accessibility analysis.* We derived measures of accessibility by foot and by bicycle to evaluate the changes resulting from delivering the intervention: see [Section 3.5](#).
6. *Underutilised capacity analysis.* We assessed the expected levels of utilisation of the additional destinations based on population catchments required for each destination and the level of density within the walkable and cyclable catchment areas: see [Section 3.6](#).

As shown in [Fig. 2](#), our analysis resulted in three key outcomes: numbers of additional destinations required to achieve the target destination standards we set for 20MNs (see [Section 4.1](#)), measures of accessibility both before and after the intervention (see [Section 4.2](#)), and measures of underutilisation for new destinations (see [Section 4.3](#)).

Our spatial and statistical analyses were conducted using R programming language ([R Core Team, 2024](#)), including the use of igraph ([Csárdi et al., 2024](#)) for network routing. Analysis was conducted on a road network derived from OpenStreetMap using methods based on those described in [Jafari et al. \(2022\)](#), using only roads and paths available for walking or cycling when measuring walking and cycling distances respectively (for example, excluding motorways). The code used for the analysis is available at <https://github.com/jibeprject/20mnMelbourne>.

3.1. Activity centres and catchment areas

A dataset of existing AC locations in Greater Melbourne in 2020 was obtained from the Victorian Government Department of Transport and Planning, comprising commercial zone polygons including central business district areas, traditional shopping strips (i.e. high streets), shopping malls, and small groups of shops. Using estimated catchment populations contained in the dataset, and information from supporting government policy documents ([Department of Transport and Planning, 2024b; Echelon Planning, 2022](#)), we classified ACs as small, medium or large. Section 1 of the Supplementary Material contains more detail on the ACs and their classifications.

While the ACs dataset included catchment population sizes, it lacked specific catchment area data. To address this data gap, we used information from the government policy documents detailing catchment areas corresponding to various AC's sizes ([Department of Transport and Planning, 2024b; Echelon Planning, 2022](#)). A walkable catchment was defined for each AC, comprising the residential dwellings within 400 m (for small ACs) or 800 m (for medium or large ACs) road network walking distance from one or more anchor points. These anchors were supermarket centroids if the AC contained one or more supermarkets ([Gunn et al., 2017](#)) (see Section 1 of the Supplementary Material for data source) or the centroid of the AC otherwise. Supermarkets were regarded as present in the AC if located within the AC polygon itself, or within a 30 m buffer. If more than one supermarket was present in an AC, then addresses able to reach any of those supermarkets within 400 m or 800 m (as applicable) formed part of that AC's catchment.

[Fig. 3](#) illustrates the study area, ACs and AC catchments, with [Fig. 3\(a\)](#) showing the overall distribution of AC catchments within the study area, while [Fig. 3\(b\)](#) shows more detail for tightly-clustered catchments around central Melbourne, displaying both 400 m (small) and 800 m (medium and large) AC catchments.

The dataset file obtained from Department of Transport and Planning contained 1157 ACs. After excluding undeveloped ACs and those with no people within walkable distance, 1141 ACs remained for analysis. As described further in this [Section 3](#), we modelled an intervention of adding destinations to serve people living in all these AC catchments, thereby covering a large proportion of Melbourne's area and population. While the ACs have been identified by government as having potential to support 20MNs ([Department of Transport and Planning, 2024a](#)), we do not suggest that it is current government policy to apply this treatment to *all* ACs in the manner we have modelled.

3.2. Destinations and population with access

We identified 14 essential destinations needed for a 20MN within categories of retail, hospitality, healthcare, education, community

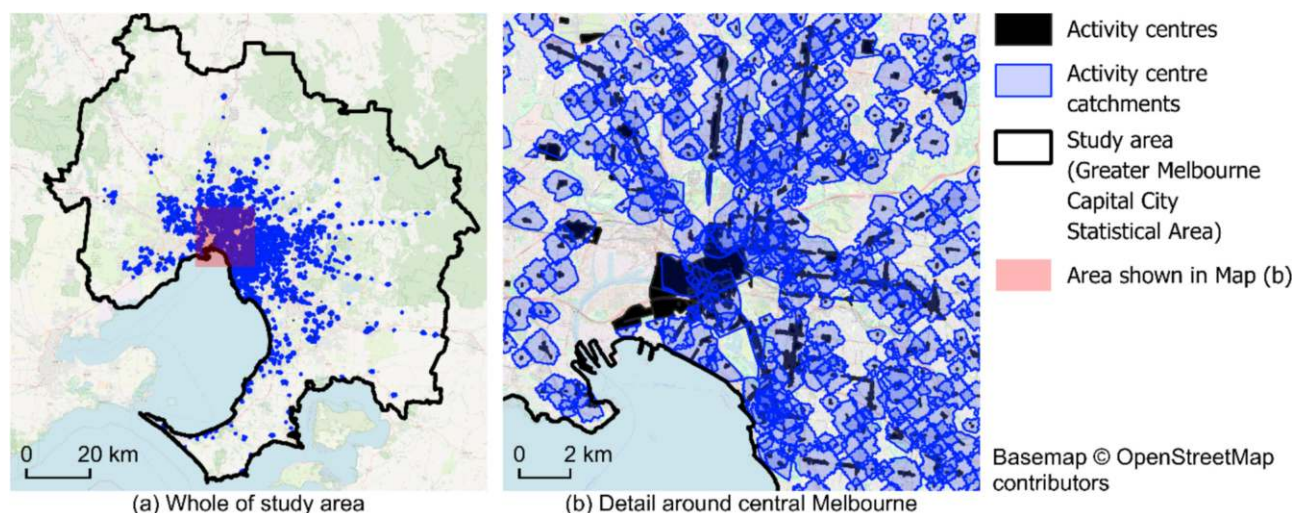


Fig. 3. Study area, small and medium/large activity centres and their respective 400 m or 800 m walkable population catchments.

services, open spaces, and public transport facilities.

Destination categories were selected based on neighbourhood-level destination types used in Victorian Government 20MNs policy documents as benchmarks for assessing ACs potential to become 20MNs (Department of Transport and Planning, 2024b; Echelon Planning, 2022; State Government of Victoria, 2019). Destination categories that typically serve an area larger than a single neighbourhood, such as secondary schools which serve multiple neighbourhoods, were not included. Walking distance on the walkable road network was set at 800 m (a 10 minute walk) for most destination categories, also consistently with those policy documents (Table 1). However, for some categories (e. g. parks) a shorter distance of 400 m was set, consistently with expectations that those destination types should be available within a 5

minute walk (Thornton et al., 2022). For bus, tram and train stop distances, we used residential development provisions used by all local governments in Victoria (ResCode clause 56-04).

Table 1 describes the destination main categories and types within each and their walking distances. We also show the equivalent cycling distances for the same travel time calculated at 12 km/h (which is, conservatively, slightly lower than the cycling speed of 13.1 km/h for Victorian journey to work trips identified by Both et al. (2022)). Walking and cycling distances were used to calculate accessibility for the base case scenario and for achieving 20MNs, building upon existing ACs.

Based on discussions with our Policy Advisory Group members and the mix of coverage thresholds of 95 %, 80 % and 70 % used by Jafari et al. (2023), the percentage target for population walking access was set at 80 %. For example, using the first two lines of Table 1, an AC meets the supermarket target if at least 80 % of the population in that AC's catchment have access to a supermarket within 800 m, and the convenience store or supermarket target if at least 80 % of the population has access to either a convenience store or a supermarket within 400 m walking distance. Similarly, an AC meets the public transport target if at least 80 % of the population in that AC's catchment have access to either a bus stop within 400 m, a tram stop within 600 m or a train station within 800 m walking distance. A target of 80 % rather than 100 % was chosen recognising that planning is often based slightly larger distances (such as 1 km), various policy standards exceed 800 m (Arundel et al., 2018), and a neighbourhood where 80 % of people have access within 800 m could still be viewed as walkable even if other people need to walk a little further. For cycling, we did not define a different set of AC catchments for the purpose of adding additional destinations as described in Section 3.4, but instead measured accessibility by bike, using the cycling distances set out in Table 1, to destinations originally intended for a walkable catchment. This decision aligns with our goal of illustrating the significance of cycling in enhancing access to essential destinations (Both et al., 2022), particularly in low-density cities like Melbourne.

3.3. Baseline assessment

Based on the identification and mapping of ACs, their catchment areas, destinations requirements for a 20MNs, and population with access to those within a walkable threshold, we began the modelling of 20MNs in Greater Melbourne by assessing the baseline level of ACs meeting the target of 80 % of people in its catchment having walking access to each destination type, using the following method:

Table 1
Destination types and walking and cycling distances.

Destinations: categories and type	Travel time	Walking distance	Cycling distance
<i>Retail, hospitality and local services</i>			
Supermarket	10 min	800 m	2 km
Convenience store or supermarket	5 min	400 m	1 km
Restaurant or café	5 min	400 m	1 km
Pharmacy	10 min	800 m	2 km
Post office	10 min	800 m	2 km
<i>Health</i>			
GP	10 min	800 m	2 km
Maternal & child health centre	10 min	800 m	2 km
Dentist	10 min	800 m	2 km
<i>Education and community</i>			
Childcare	10 min	800 m	2 km
Kindergarten	10 min	800 m	2 km
Primary school	10 min	800 m	2 km
Community centre or library	10 min	800 m	2 km
<i>Open space</i>			
Park	5 min	400 m	1 km
<i>Public transport</i>			
Bus, train or tram	5 min (bus), 7.5 min (tram) or 10 min (train)	400 m (bus), 600 m (tram) or 800 m (train)	1 km (bus), 1.5 km (tram) or 2 km (train)

1. Residential addresses within Greater Melbourne were identified.
2. Using census data providing counts of population and dwellings, each address was weighted to represent a number of people and a number of dwellings.
3. Each AC's catchment was determined as the addresses able to reach an anchor point (i.e. supermarket or centroid, as described in Section 3.1) within a 400 m (for small ACs) or 800 m (for medium and large ACs) network distance.
4. For each AC and destination type, the road network distance from each address to its nearest destination of each type was determined.
5. An AC was considered to meet the target for a destination type if at least 80 % of people in the AC catchment were within the walking distance for that type of destination specified in Table 1.

Section 2 of the Supplementary Material provides additional detail about the baseline assessment process.

Section 3 of the Supplementary Material describes two alternative methods also considered for conducting the baseline assessment, and explains the rationale for selecting the chosen method as providing a primary focus on people as the users of destinations, and on anchors as relevant central locations within ACs. Section 4 of the Supplementary Material provides a breakdown by population size of ACs meeting the 80 % target.

3.4. Creating 20MNs by adding destinations

Where an AC did not meet the target for a destination type at the baseline, notional locations for the placement of additional destinations were determined. This was achieved by using the following method to identify locations that would provide access for the 80 % target of people within the AC catchment area:

1. Addresses within the AC catchment area not having access to a given destination as per Table 1 ('inaccessible addresses') were identified.
2. Candidate network nodes were selected for the location of new destinations, generally being all nodes within the required walking distance of the inaccessible addresses, but preferencing nodes within the AC itself for certain destination types.
3. Network distances from the candidate nodes to the inaccessible addresses were calculated.
4. A new destination of the relevant type was notionally imagined at each candidate node and the number of additional people (determined by the population weights of the inaccessible addresses)

served within its catchment distance was calculated; the candidate node that maximised the number of people additionally served was then selected as the notional location of the new destination.

5. Each AC was re-assessed, using the same procedure as for the baseline assessment but taking account of the new destination location. If the AC still did not meet the target, a further notional location was determined using the same procedure and iterated until the 80 % target was met for that destination type.

Fig. 4 illustrates the process, and Section 5 of the Supplementary Material provides additional detail.

This placement process is not intended to assert that the selected locations are optimal or realistic locations for the destinations. The concept of building out the ACs is realistic, and follows from them having been identified by policymakers as activity centres for development, but the specific street locations selected by the placement process may not be available. Rather, this paper is intended as a proof of concept with the objective of modelling outcomes that would arise if the target numbers of people had access to destinations of the relevant type, and the purpose of the placement process is to adopt a parsimonious approach to determining the number of new destinations that would be required to achieve the target. This allows the magnitude of the task of building out the ACs to be shown (see Table 3) without inflating it by adding more destinations than necessary to achieve the target.

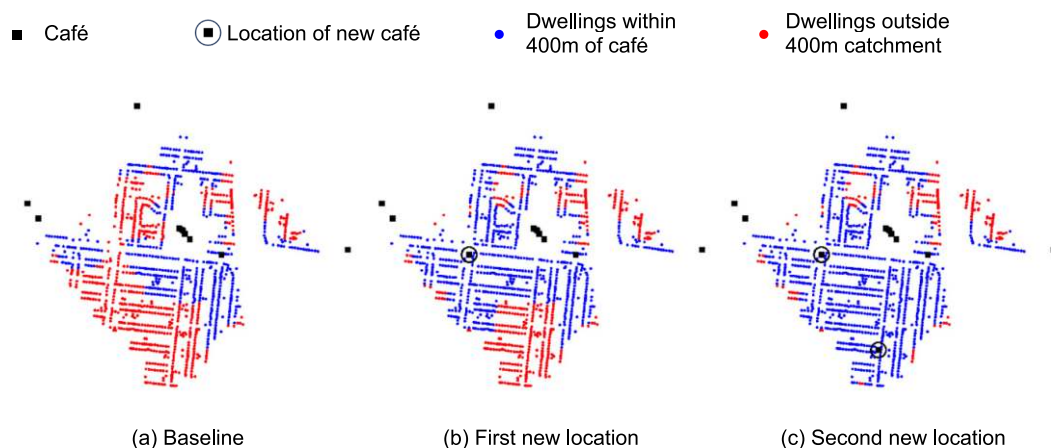
3.5. Accessibility analysis

Accessibility analysis was undertaken for both walking and cycling. An accessibility score for each address within Greater Melbourne was calculated by allocating one point for each destination type accessible within the required walking or cycling distances from a given address, as set out in Table 1, for a maximum possible score of 14. For presentation purposes, we calculated accessibility scores for LGAs at baseline and after adding the new destinations to achieve 20MNs around ACs (we referred to this as the intervention).

The LGA score for each of the baseline and intervention was calculated as:

$$LGA\ score = \frac{\sum_{i=1}^n (AS_i * PW_i)}{\sum_{i=1}^n PW_i} \quad (1)$$

where AS_i is the score for each address i in the LGA and PW_i is the



Notes: (a) Baseline: 50% of people are within 400m walking distance of a café, and so a new café location is required. (b) A new location is determined, resulting in 72% of people being within 400m walking distance of a café, and so a further new café location is required. (c) A second new location is determined, resulting in 87% of people being within 400m walking distance of a café, and so the AC meets the target.

Fig. 4. Illustration of process for adding new destination locations to an AC.

population weight for the address, as calculated using Eq. S1 in Section 2 of the Supplementary Material.

The difference for each LGA was calculated as

$$\text{difference} = \text{LGAScore}_{\text{int}} - \text{LGAScore}_{\text{base}} \quad (2)$$

where $\text{LGAScore}_{\text{base}}$ and $\text{LGAScore}_{\text{int}}$ are the LGA scores allocated for the baseline and intervention respectively.

For ease of comparison, the 31 LGAs that comprise the majority of the study area were grouped into inner (within 10 km of the central business district (CBD)), middle (10–20 km of the CBD) and outer (over 20 km of the CBD), as shown in Fig. 5.

Alternative methods of scoring, where destinations were assigned weights (e.g. a supermarket has a weight of 5 and a café a weight of 1; multiple cafés have incremental weights), were also considered and compared. These are described in Section 6 of the Supplementary Material. The alternative methods did not produce notably different results. The results presented here are aggregated by LGA, however calculation of scores at the address level means they could also be aggregated for other spatial units as required, using the same method.

3.6. Underutilised capacity analysis

We measured the extent to which new destinations placed in and around ACs met target levels of access to achieve 20MNs and whether this would result in underutilisation of those destinations by residents within a walkable catchment. Underutilisation was based on each destination's population catchment requirements (see Table 2) and observed density for the ACs' catchment areas. Section 9 of the Supplementary Material presents the results of the same analysis for existing destinations. The analysis was undertaken for both walking and cycling.

For this analysis we: (1) allocated a 'population requirement' to each destination type; (2) calculated a utilisation viability score for each new destination, by comparing the number of people served by the destination to its population requirement; and (3) aggregated the utilisation viability scores for each LGA.

The population requirements are set out in Table 2, based on policy documents for allocation of new facilities and prior literature (Australian Social & Recreation Research Pty Ltd, 2008; Department of Transport and Planning, 2024b; Jafari et al., 2023) as specified in the Table, supplemented in cases where not specified in policy documents and literature by discussions with practitioners and policymakers in our

Table 2

Population requirements for destination types.

Destination type	Population requirement	Policy document and literature sources
<i>Retail, hospitality and local services</i>		
Supermarket	10,000	Department of Transport and Planning (2024b); Jafari et al. (2023)
Convenience store	1,000	Jafari et al. (2023)
Restaurant or café	1,000	-
Pharmacy	10,000	-
Post office	10,000	-
<i>Health</i>		
GP	1,000	Jafari et al. (2023)
Maternal & child health centre	16,000	Australian Social and Recreation Research Pty Ltd (2008); Department of Transport and Planning (2024b)
Dentist	1,000	-
<i>Education and community</i>		
Childcare	9,000	Australian Social and Recreation Research Pty Ltd (2008); Jafari et al. (2023)
Kindergarten	10,000	Australian Social and Recreation Research Pty Ltd (2008)
Primary school	9,000	Australian Social and Recreation Research Pty Ltd (2008)
Community centre	8,000	Australian Social and Recreation Research Pty Ltd (2008)
<i>Open space</i>		
Local park	1,000	Australian Social and Recreation Research Pty Ltd (2008)
<i>Public transport</i>		
Bus stop	1,000	-

Policy Advisory Group, and by analogy with similar destination categories.

The utilisation viability score for each new destination was calculated, for both walking and cycling distances, as:

$$\text{utilisation viability score} = \frac{\text{people served}}{\text{population requirement}} \quad (3)$$

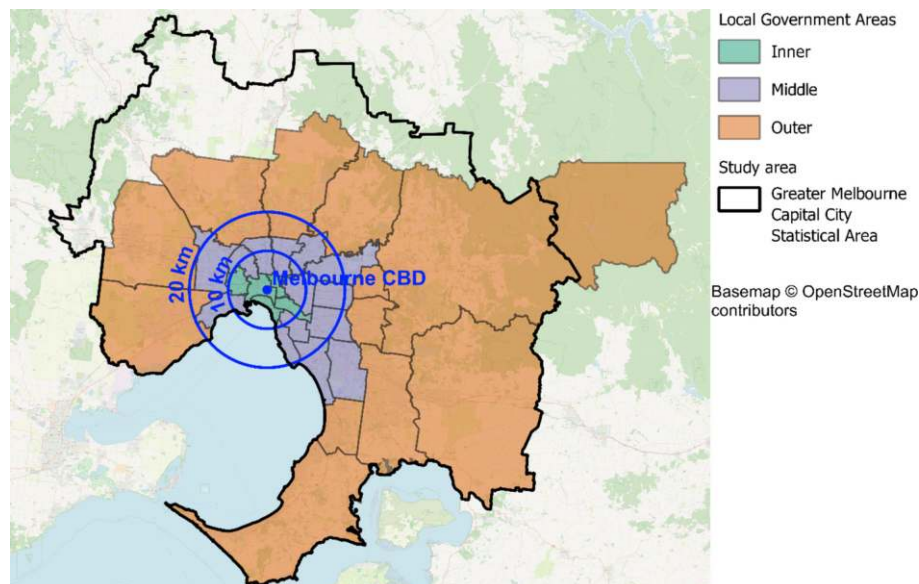


Fig. 5. Classification of local government areas as 'inner', 'middle' and 'outer', based on distance from central business district (CBD).

Table 3

Number of ACs meeting, and not meeting, the target for each destination type in the baseline assessment, and number of new locations added to meet target.

	ACs where at least 80 % of people are within specified walking distance of destination type in baseline assessment		ACs not meeting target in baseline assessment		No of new destinations required to be added across the study area
	No.	%	No.	%	
Accessible within 800 m walking distance					
Supermarket	506	44	635	56	541
Pharmacy	323	28	818	72	690
Post office	169	15	972	85	848
GP	570	50	571	50	492
Maternal & child health centre	150	13	991	87	828
Dentist	419	37	722	63	623
Childcare centre	906	79	235	21	210
Kindergarten	323	28	818	72	654
Primary school	419	37	722	63	599
Community centre or library	126	11	1015	89	866
Accessible within 400 m walking distance					
Convenience store or supermarket	208	18	933	82	1234
Restaurant or café	281	25	860	75	1359
Park	616	54	525	46	554
Accessible within 400 m (bus), 600 m (tram) or 800 m (train) walking distance					
Bus stop, tram stop or train station	930	82	211	18	259

Note: total number of ACs is 1141.

where *people served* is the sum of the population weights for the addresses within the required walking or cycling distance (as set out in Table 1) of the new destination and *population requirement* is the applicable population requirement from Table 2.

That is, the utilisation viability score is the number of people living in the walking or cycling catchment of the destination (based on baseline population), as a fraction of the number required for viability. We have used the term ‘utilisation viability score’ as the score is a measure of the viability of the destination (that is, assessing whether it will have enough people in its catchment for its viability), but with ‘viability’ qualified by ‘utilisation’ to acknowledge that the score assesses only the potential utilisation aspect of viability. The score does not assess other aspects of viability such as the availability of land, the cost of doing business in the relevant locality, or the quality of the services provided, all of which would affect whether or not the destinations were used, and are matters meriting further investigation in the concept of x-minute cities. As discussed further in Section 5.4, the score is an assessment of whether a destination’s surrounding area is able to supply the destination with sufficient utilisation on the basis of active transport, not whether it would be viable taking account of all transport modes.

The utilisation viability score for each LGA and destination type was calculated as the mean of the utilisation viability scores for new destinations placed in that LGA.

The utilisation viability scores for each LGA were also compared with the dwelling density of the ACs catchments in the LGA. For this purpose, a single utilisation viability score was calculated for each LGA as the mean of the LGA’s utilisation viability scores for all destination types, and the AC catchment density for each LGA was calculated as:

$$AC \text{ catchment density} = \frac{\sum_{i=1}^n (DW_i)}{AC \text{ area}} \quad (4)$$

where DW_i is the dwelling weight for each address i located within an AC catchment and within the LGA as calculated using Eq. S2 in Section 2 of the Supplementary Material, and $AC \text{ area}$ is the combined area (without double-counting, in the case of overlaps) of the AC catchments, represented as polygons, located within the LGA.

4. Results

4.1. Baseline assessment and additional destinations required to achieve the 20MN

Table 3 shows the results of the baseline assessment, indicating the number and percentage of ACs that did not meet the target of 80 % of people being within the required walking distance for each destination type. It also shows the number of ACs requiring additional destinations, and the number of new destinations required for each AC to meet the target. The latter provides an assessment of the destinations required to achieve the 20MNs.

As can be seen in Table 3, fewer than 50 % of ACs met the 80 % target of access to most destination types within a walkable catchment, with childcare centres, parks and public transport stops being the exceptions.

Where destinations were required to be within an 800 m walking distance, the number of new destinations required to be added was less than the number of ACs requiring them. This is because overlapping ACs mean that a new location which is placed to help one AC meet its 80 % target may also help another nearby AC meet the target. However, where the required walking distance was 400 m, the number of new destinations required exceeds the number of ACs not meeting the target in the baseline, because some large and medium ACs with 800 m catchments require more than one new destination in order to achieve the required percentage of people to be within 400 m of each of these destination types.

4.2. Accessibility analysis

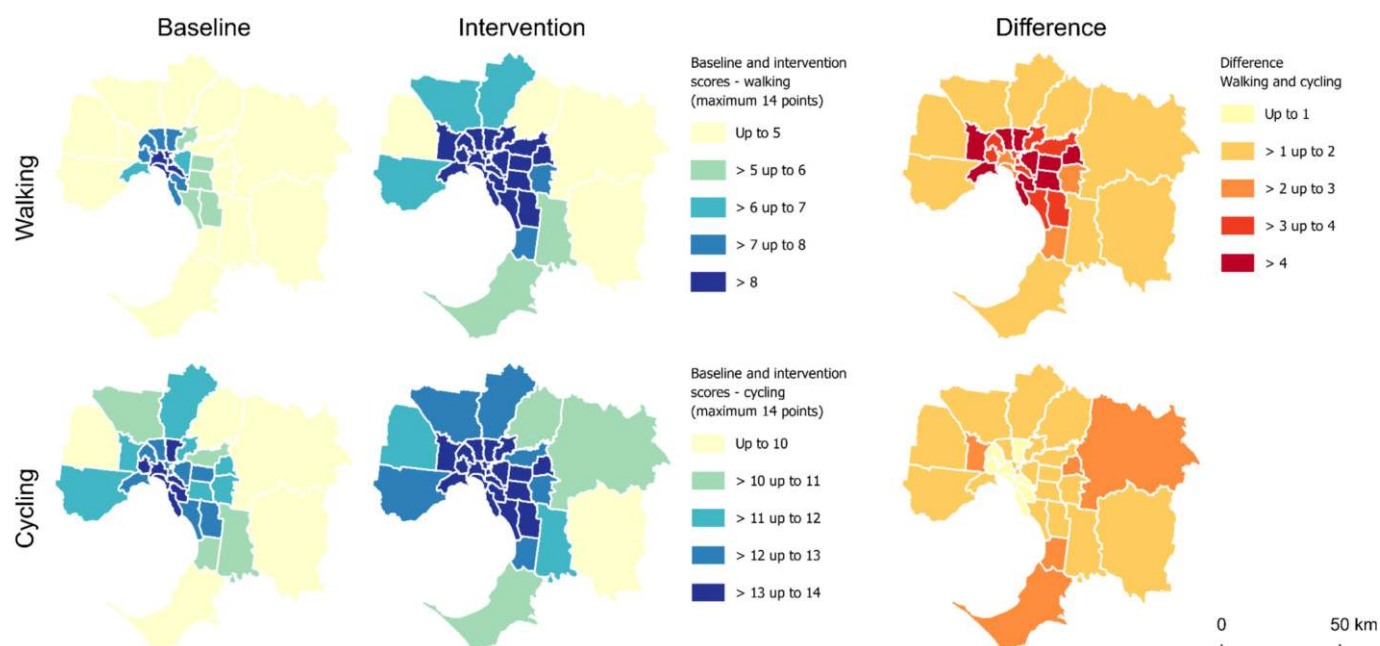
Table 4 shows the LGA walking and cycling accessibility scores, which are also illustrated in Fig. 6. Section 6 of the Supplementary Material sets out additional results using alternative scoring methods. The ‘rank’ shown in Table 4 is the ranked ordering of walking or cycling accessibility score differences, where ‘1’ shows the LGA where the intervention produces the highest change in score.

The walking results for both baseline and intervention show the highest accessibility scores in inner and middle LGAs, and lowest in outer LGAs. The differences between the baseline and intervention scores are generally highest in the inner and middle LGAs (and generally

Table 4

Walking and cycling accessibility scores (maximum =14) for each Local Government Area, showing the scores for baseline, intervention, the difference between them, and the ranked differences.

Local Government Area		Scores: Baseline (Base), Intervention (Int), Difference (Diff) and rank of difference (Rank)							
		Walking				Cycling			
		Base	Int	Diff	Rank	Base	Int	Diff	Rank
Inner	Stonnington	8.5	12.4	3.9	11	13.3	14.0	0.7	27
	Yarra	9.5	13.1	3.7	13	13.0	14.0	1.0	23
	Maribyrnong	7.7	10.8	3.1	18	13.3	14.0	0.7	28
	Melbourne	10.1	13.1	2.9	19	13.8	14.0	0.2	30
	Port Phillip	10.2	13.0	2.9	21	13.9	14.0	0.1	31
Middle	Whitehorse	5.7	11.2	5.5	1	12.3	14.0	1.7	13
	Glen Eira	7.1	12.2	5.1	2	13.1	14.0	0.9	25
	Darebin	7.4	12.3	4.9	3	13.1	13.9	0.8	26
	Merri-bek	7.3	12.2	4.9	4	12.7	14.0	1.2	21
	Boroondara	7.0	11.8	4.8	5	12.5	14.0	1.4	17
	Monash	5.3	10.1	4.8	6	11.9	13.8	1.9	7
	Hobsons Bay	6.4	10.8	4.4	7	12.5	13.9	1.4	19
	Brimbank	4.9	9.1	4.2	8	11.2	13.6	2.4	2
	Bayside	7.1	11.3	4.2	9	13.4	14.0	0.6	29
	Banyule	5.8	9.5	3.7	12	11.8	13.6	1.8	8
	Kingston	5.9	9.5	3.6	14	12.3	13.6	1.3	20
	Moonee Valley	7.1	10.6	3.5	15	12.9	13.8	0.9	24
	Greater Dandenong	5.5	8.9	3.4	16	12.3	13.4	1.1	22
	Manningham	4.7	8.0	3.4	17	10.9	12.8	2.0	6
	Maroondah	4.5	8.6	4.1	10	11.0	13.4	2.4	3
Outer	Frankston	4.4	7.2	2.9	20	10.7	12.8	2.0	4
	Knox	4.6	7.0	2.4	22	11.2	12.9	1.7	12
	Mornington Peninsula	3.0	5.0	2.0	23	8.4	11.0	2.5	1
	Whittlesea	4.8	6.8	2.0	24	11.3	12.7	1.5	16
	Hume	4.5	6.4	1.9	25	10.4	12.2	1.8	11
	Yarra Ranges	3.2	5.0	1.7	26	8.9	10.9	2.0	5
	Wyndham	4.4	6.0	1.7	27	11.0	12.4	1.4	18
	Casey	4.1	5.5	1.5	28	10.3	11.8	1.5	14
	Melton	3.6	4.7	1.1	29	9.3	11.1	1.8	10
	Cardinia	3.2	4.3	1.1	30	8.5	9.9	1.5	15
	Nillumbik	3.2	4.2	1.1	31	8.7	10.5	1.8	9

**Fig. 6.** Walking and cycling accessibility scores for local government areas.

higher for middle LGAs than inner), and lowest in the outer LGAs.

The results suggest that the effect of increasing access to destinations would improve access for people in LGAs with the highest baseline results. This is because the ACs in the inner and middle suburban LGAs are more closely clustered, and hence the overall accessibility of these ACs'

catchments will be enhanced with the addition of new local destinations. Conversely, in the outer suburban LGAs, the ACs are spread apart, and therefore investment in or around one AC would have little benefit for another AC catchment.

Cycling results show a similar pattern of higher baseline and

intervention scores in inner and middle LGAs than in outer LGAs. The inner LGAs and some middle LGAs scores approach the maximum available points for access to destinations in the baseline, reflecting an ability to reach all destination types within 10 minutes (or less) by bicycle. However, most notably, the differences are generally higher in the outer LGAs than inner and middle LGAs, contrary to the pattern observed for walking results, indicating that the intervention raises scores more in outer LGAs, although usually not enough to enable ACs to achieve the maximum accessibility score. Part of the reason why the intervention raises scores less in inner and middle LGAs in inner and middle LGAs is that some of those LGAs already had scores near the maximum available level. Nevertheless, the results show that more outer LGA residents could benefit from the improvements made in and around ACs if cycling was enabled by safe cycling infrastructure, because the longer cycling distances increases access within 10 minutes.

This is shown graphically in Fig. 6, where the 'baseline' and 'intervention' maps for both walking and cycling show a pattern of highest accessibility scores (dark blue) in the inner LGAs; while the 'difference' maps for walking show the highest levels of change in inner and middle LGAs, and the 'difference' maps for cycling show the highest levels of change in outer LGAs.

Looking at accessibility across Greater Melbourne as a whole, we found that only 0.5 % of households (0.4 % of people) across Greater Melbourne live within the required walking distance (as shown in Table 1) of all 14 destination types under baseline conditions, while the intervention increases this to 29.3 % of households (26.8 % of people). Table S10 in Section 7 of the Supplementary Material sets out a more detailed breakdown of households and people, and also shows the applicable scores when cycling distances are used instead of walking distances.

4.3. Underutilised capacity analysis

As noted earlier, each type of destination has a different 'population requirement' (see Table 2), representing the number of people required in the destination's catchment to support its viability. Hence adding a new destination to achieve the 20MN target of 80 % of residents having spatial access with 800 m (or 400 m, where applicable for the relevant destination type as specified in Table 1) may mean that destinations within a walkable catchment could be underutilised, depending on the area's population density. Figs. 7 and 8 respectively show the walking and cycling utilisation viability scores for each LGA and new destination type. Tables S11 and S12 in Section 8 of the Supplementary Material set out the details of the scores displayed in those Figures in table format.

As Fig. 7 illustrates, for walking, a small number of destination types (GP and dentist) have a utilisation viability score of over 0.75 in all LGAs after intervention, indicating that an additional destination of those types placed to meet the 80 % target would, on average, be fully utilised within the destination's walkable catchment, or nearly so, across all LGAs. However, most other types of destination types only reach or approach full utilisation in inner LGAs, with considerably lower utilisation viability scores achieved elsewhere. While most inner LGAs reach full utilisation within a walkable catchment for many destination types, the outer eastern LGAs of Cardinia and Yarra Ranges had utilisation viability scores of 0.25 or below for 8 out of the 14 destination types studied. This reflects the low population densities observed in these areas, making it infeasible to add new destinations to achieve access within 800 m.

However, as illustrated in Fig. 8, if the catchments for new destinations are increased from a walking distance of 400 m or 800 m to a cycling distance of 1 km or 2 km (as applicable for the relevant

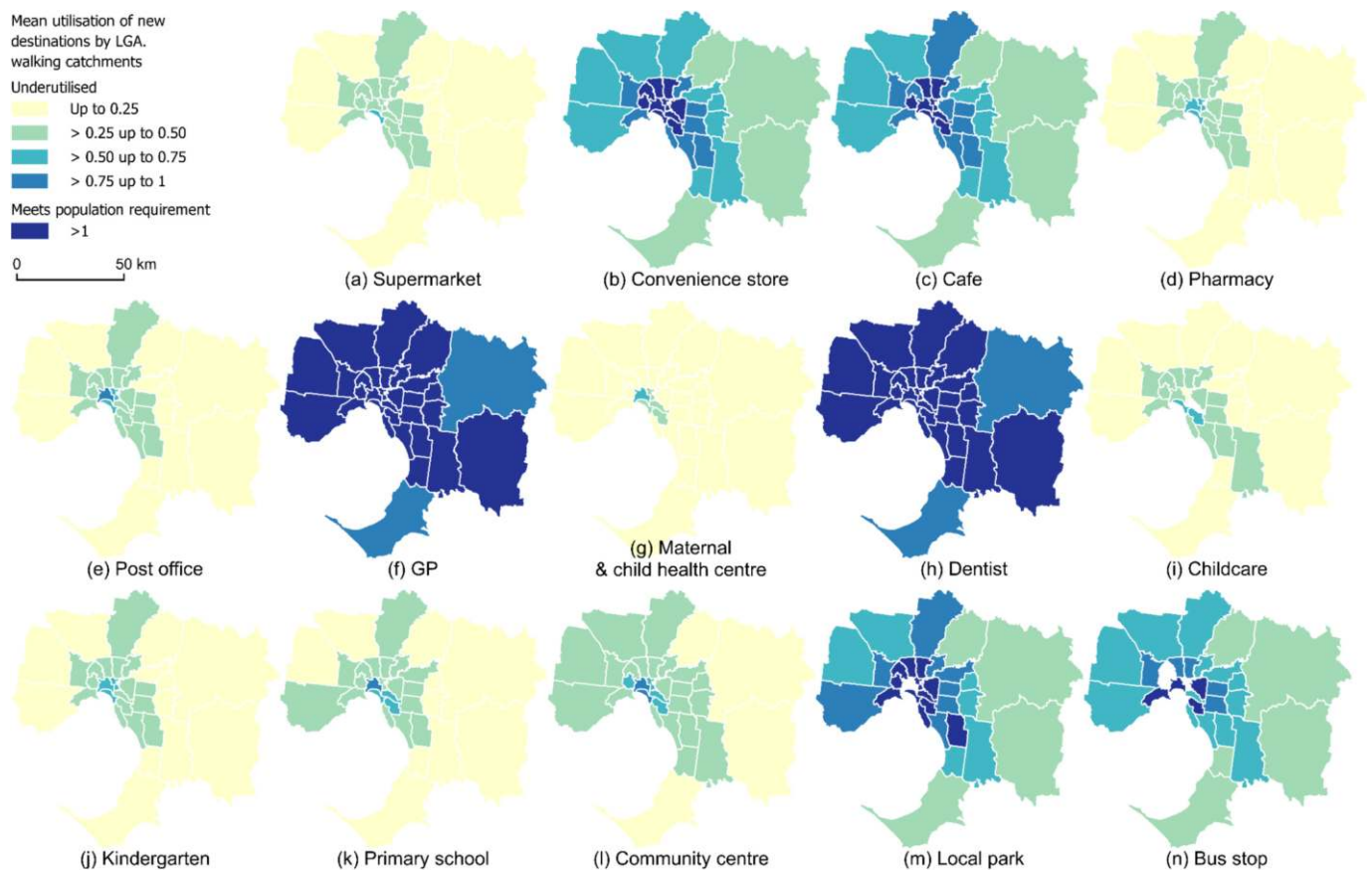


Fig. 7. Utilisation viability scores (number of people living in the destination catchment as a fraction of the number required for viability) for new destinations placed to meet target access level, based on people within walking distance catchments, by destination type and local government area (note: where no destinations of a particular type were placed in a local government area, that local government area is omitted from the map for that destination type).

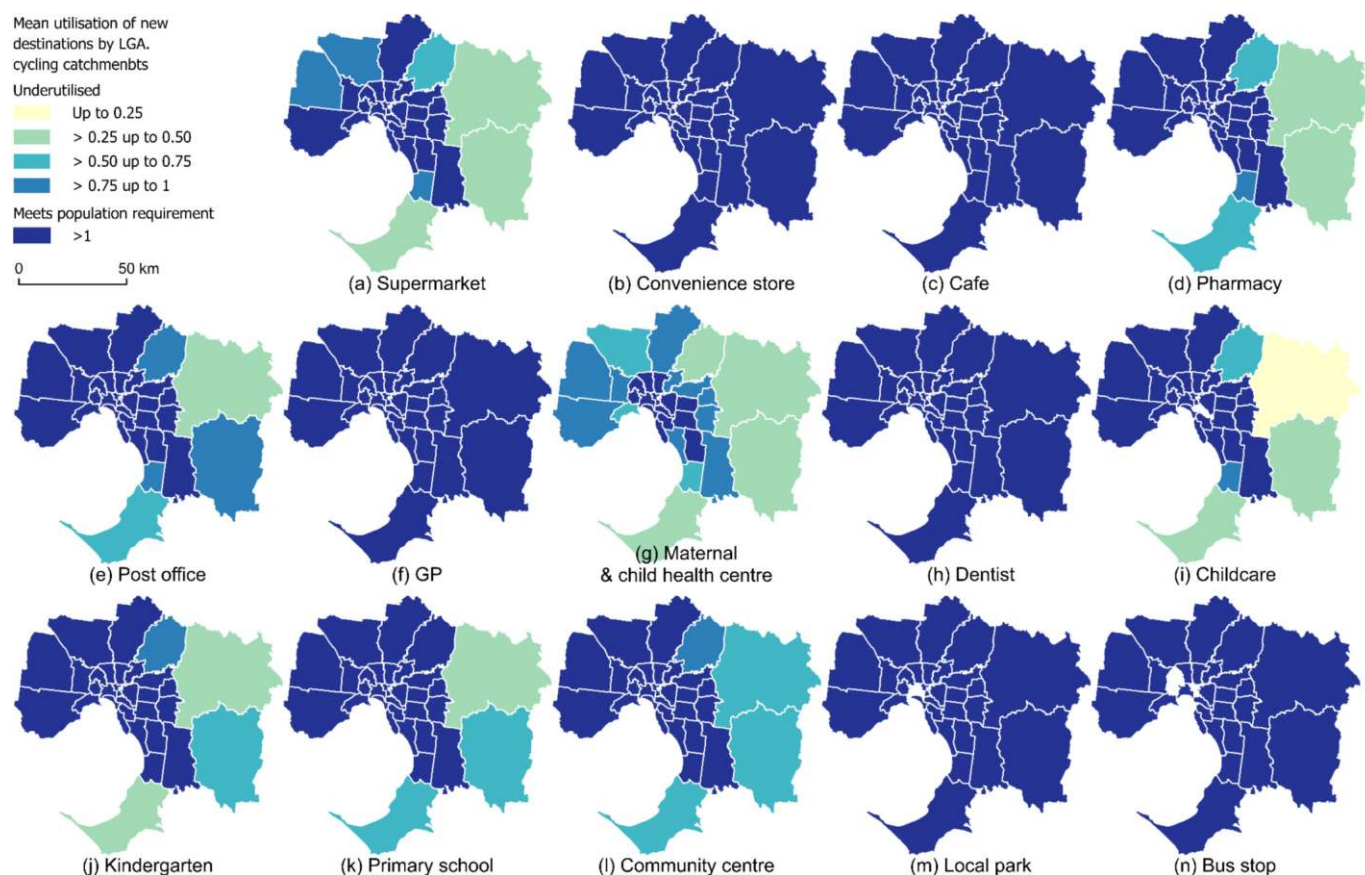


Fig. 8. Utilisation viability scores (number of people living in the destination catchment as a fraction of the number required for viability) for new destinations placed to meet target access level, based on people within cycling distance catchments, by destination type and local government area (note: where no destinations of a particular type were placed in a local government area, that local government area is omitted from the map for that destination type).

destination type, as set out Table 1), most destination types are fully utilised in most LGAs. Even in the outer LGAs, most utilisation viability scores are at least 1 (full utilisation) in most LGAs, though three LGAs (Cardinia, Mornington Peninsula and Yarra Ranges) still have utilisation viability scores below 0.75 for at least half of the destination types within their walkable or cyclable catchments.

Dwelling density is a major driver of the results observed in Figs. 7 and 8.

This is apparent in Fig. 9, which plots the mean walking and cycling utilisation viability scores for the LGAs against the density, in dwellings per hectare, of AC catchments in the LGA. While the range in mean cycling scores (0.3 to 1.5) is smaller than the range in walking scores (1.2 to 9.2), the plots show the strong association between density and both walking and cycling utilisation viability scores, with higher mean utilisation viability scores corresponding to the higher densities found in the inner LGAs.

5. Discussion

5.1. Modelling the benefits of x-minute city policies in low-density settings

Our study appears to be the first natural experiment study designed to evaluate the potential to deliver an x-minute policy. Our study provides novel and generalisable methods for quantifying impacts on both accessibility and behaviour that can be applied elsewhere. By providing a specific example of implementation, we demonstrate how quantification can be achieved, extending both to the accessibility benefits of the scenario, and to the consequences for utilisation of infrastructure.

Importantly, while our study applied our methods to Melbourne, the methods are adaptable for use in other locations worldwide. A city

wishing to implement an x-minute city policy will need to consider both population and destinations. In relation to destinations, which is the focus of this paper, the city will need to determine the objectives of the policy (what level of access should be achieved and by how many people), measure the current position (baseline), determine the deficit (what needs to be added to achieve the policy), and assess the viability of what needs to be added. Our study contributes to the literature both by providing an example of a natural experiment study designed to quantify the potential for delivery and benefits of an x-minute policy in a low-density context, and by providing generalisable methods for such quantification which can be used elsewhere.

Employing our methodological framework in the context of existing ACs, we found significant benefits associated with achieving 20MNs in established areas.

We demonstrated how the 20MN scenario can improve accessibility by foot and bicycle to essential destinations for daily living. Meeting the target of 80 % of people in each AC catchment being within the required walking distance of each destination type required the counterfactual addition of new daily living amenities and services. Overall the intervention would result in 29.3 % of households in Greater Melbourne having access to all 14 destination types, an increase from 0.5 % at baseline. However, these benefits would not be uniformly distributed across the city.

The greatest walking accessibility gains were in inner and middle city areas, while cycling accessibility was most improved in the outer suburban areas. Two primary factors account for these outcomes: firstly, the dense concentration of ACs in inner and middle suburban areas, which benefit from a higher baseline level of amenities in comparison to outer areas; and secondly, the increased distances reachable by cycling compared with walking. In inner LGAs, ACs and their catchments cover

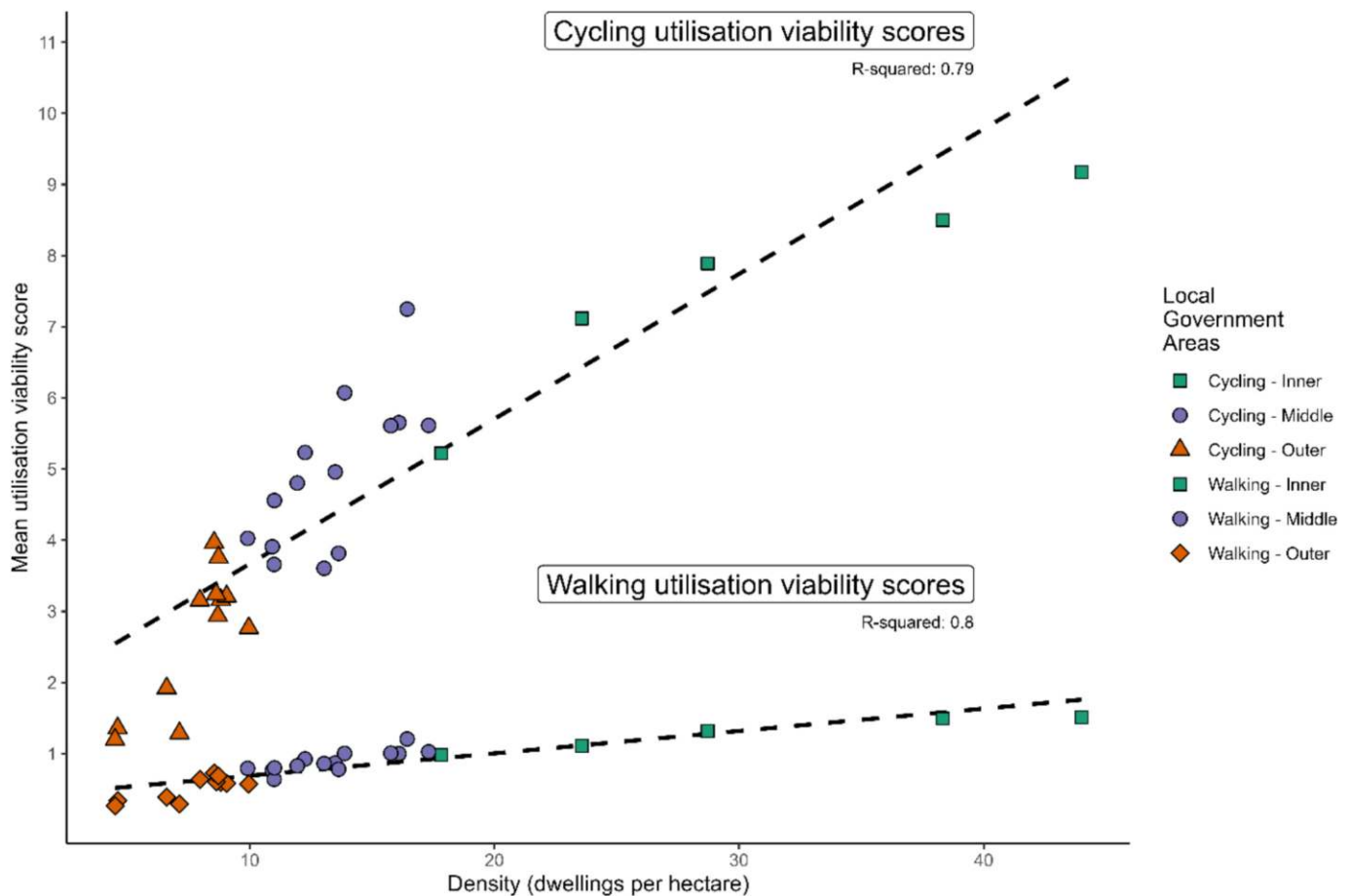


Fig. 9. Mean walking and cycling utilisation viability scores (number of people living in the destination catchment as a fraction of the number required for viability) for new destinations for LGAs compared to dwelling density, where a score of 1 represents full utilisation and so scores below 1 represent underutilisation.

most of the LGA, with catchments often overlapping, and are often well-supplied with a baseline level of destinations. The dense spacing means that the intervention results in almost the whole of these LGAs being well-supplied with destinations, and modest increases in good baseline accessibility scores result in high intervention accessibility scores. In middle LGAs, baseline accessibility scores are lower, particularly for walking; but AC catchments cover much of the populated parts of these LGAs, and so the intervention targeted at the ACs is able to achieve good overall increases in walking accessibility scores across these LGAs. Whereas in outer LGAs, more of the population resides outside AC catchments, and so the intervention which builds out these catchments still leaves much of the population outside the reach of walking access to the destinations. However, many of the people living outside the AC walkable catchments are within cycling distance of the ACs, reflected by the good increases in cycling accessibility scores for outer LGAs. This highlights the importance of cities considering improvements in cycling infrastructure even in outer suburban areas to enable active transport and reduce transport inequities.

Our finding that baseline accessibility scores are highest in inner areas, and lowest in outer, is intuitive in regard to Melbourne's geography and urban structure, and consistent with the findings of Thornton et al. (2022) for Melbourne and Adelaide, and Weng et al. (2019) for Shanghai. Our finding as to the spatial distribution of benefits of the intervention is also as expected.

5.2. Challenges and mitigations in low-density environments

However, our research highlights the potential challenges in achieving x-minute neighbourhoods in cities characterised by low

density and extensive urban sprawl. The outcomes of the utilisation capacity analyses indicate that implementation of an x-minute concept in a low-density city, without concurrent increases in population density, may prove unfeasible. Our analysis of underutilisation patterns within walkable or cyclable catchments highlights this point, showing likely significantly higher levels of underutilisation by residents living within walkable catchments in outer city areas, which exhibit the lowest population densities, indicating the need to supplement x-minute policies with increases in density.

Nevertheless, we observe a mitigating effect on underutilisation when cycling accessibility is considered, suggesting that investment in safe cycling infrastructure could play a crucial role in enhancing the viability of x-minute policies in sprawling urban contexts. In the Melbourne example, our sensitivity testing regarding the catchment areas of ACs demonstrates that expanding the catchment of these centres, such as extending catchments from 800/400 m to 1 km or 1.2 km, leads to notable increases in the potential utilisation rates of all the destinations necessary to deliver 20MNs by residents using active modes of transport within the catchment. This indicates the need when considering or implementing x-minute policies in low-density cities to include provision for cycling, including the availability of adequate cycling infrastructure and appropriate speed limits for cycling safety.

As explained in Section 3.4, the modelling of additional destinations and their placement at specific locations is not intended to imply that those are optimal, or even possible, locations. In practice, unavailability of modelled sites would mean even more destinations than the numbers shown in Table 3 would be required for the ACs to achieve 20MN status. There are reasons – such as market forces, zoning or government planning priorities – why destinations do not already exist, whether at the

modelled locations or elsewhere, and the baseline utilisation viability scores for many new destinations reflect these commercial and government outcomes. That is, the baseline scores reflect (among other things) the impracticability of providing some of these destinations under current conditions, leading us to conclude that increases in density, alongside plans for provision of greater amenity, are needed if 20MNs are to be achieved. Under baseline conditions, travel to destinations not reachable by walking or cycling must be achieved by public transport or (more frequently) car, or else the trip cannot be taken. The x-minute concept, as described in the literature referred to in [Section 2.1](#), aims to achieve a different urban structure, giving people the option to reach daily destinations without using cars; but public transport is not inconsistent with the concept. A future extension of our work could include public transport, by defining ‘public transport catchments’ for ACs in place of the walking catchments used in this paper.

Our findings in this study are aggregated to be expressed at LGA level. However, our methods calculate accessibility scores for each address, and utilisation viability scores for each new destination, and these scores can be aggregated for other spatial units are required. Such aggregation could be used to prioritise ACs for development, by identifying those where the intervention achieves the highest level of improvement. Our methods are adaptable to other cities, where the following data is available: (1) target areas for locations (corresponding to the ACs used in this study), (2) road network, (3) address points with associated population data, (4) locations of existing destinations, and (5) spatial units for aggregation. The choice of destination set may vary depending on local conditions. For example, our destination set did not include secondary schools or hospitals, which in the Melbourne context would be expected to serve a much larger area than a walkable neighbourhood; but different conditions in other cities may merit their inclusion, having regard to the specific policies and context of those cities. Similarly, our selection of population requirements for destination types ([Table 2](#)) was informed by local context, and different numbers may be appropriate in different cities.

Our methods treat destinations as dynamic, modelling the addition of new destinations, but treat housing as static. Future extensions to the methods could include also making housing dynamic, allowing the modelling of scenarios where both the destinations and housing locations are adjusted.

5.3. Policy implications

Policy implications of our findings include the following.

1. While the accessibility improvements available from implementation of x-minute policies are significant, they are unlikely to be realised by increasing the density of destinations alone. Integrated planning is required. Accordingly related policies must also address both housing density and commercial development around identified ACs. However, the success of these policies depends on delivering the infrastructure required. Adequate budgets must be allocated to achieve this.
2. This should be supplemented by ensuring adequate supplies of other destinations which require more space, such as kindergartens and primary schools, close to the ACs. Planning involvement in curation of both commercial and social infrastructure will be required to deliver vibrant 20-minute neighbourhoods.
3. The scale of the additional destinations required to achieve x-minute cities in a low-density context is vast. This may not be possible for a city as a whole given the cost of retrofitting. Instead, specific areas within the city may need to be prioritised for increased density of both population and destinations. The methodology we have developed can be applied to indicate those areas which most fall short in terms of destinations, housing, or both, and therefore those which can most readily be improved to achieve targets.

4. Recognising that it will not be possible to build out the entire urban area of a low-density city into walkable neighbourhoods, cycling must be adopted as part of the x-minute city policy framework, implementing infrastructure and safety improvements (such as speed limit reductions on residential streets and cycle paths) to provide for people within an x-minute cycling distance as well as an x-minute walking distance. This not only has the potential to increase active transport, but will also assist in reducing transport inequities.

5.4. Limitations

The methods used were subject to some limitations.

Existing destination locations were based on OpenStreetMap or other data sources available to us which in some cases might both overstate and understate the existence of actual destinations of the relevant type. While OpenStreetMap has been described as state-of-the-art for street network analysis ([Boeing et al., 2022](#)) and locations of amenities and services ([Psyllidis et al., 2022](#)), such descriptions are given subject to caveats of spatially patterning in data quality, with poorer quality representation often in lower population and deprived areas towards the urban fringe. For locations where presence of destinations was underestimated, baseline accessibility would be higher, such that the number of new destinations required to meet accessibility targets would be lower than our findings suggest.

Address points and destinations were treated as located at their nearest network nodes (rather than at precise points in the middle of network links where they are located), which means that results for specific locations are approximations. This was done for computational efficiency. However, we would not expect the combined effect of those approximations to be so methodically biased as to produce overall higher or lower accessibility or utilisation results.

The locations of new destinations that formed the intervention were determined programmatically to maximise people having access to each new destination. This did not take account of other considerations (such as zoning, existing land use, local politics, community opinion, or cost) that would affect location selection. Hence, the results reflect a modelled outcome of the position that would apply if the 80 % targets for accessibility were met rather than a recommendation as to precise locations of new infrastructure.

The measure used for assessing utilisation viability is a simple distance-based measure, based on the number of people residing in a catchment area. A focus on distance does not address other factors that would influence actual utilisation (such as attractiveness of other nearby destinations, or convenience in relation to linked trips), and the measure does not address the ways in which active transport access constraints are practice resolved by public transport or (most commonly, in Melbourne) car travel. We determined the use of the distance-based measure to be desirable in our context because of our focus in this research on active transport, and justified because it achieves our objectives of comparing utilisation viability (in this sense) between different areas, and measuring the extent to which destination availability is consistent with the achievement of an x-minute city concept in a low-density city such as Melbourne. The utilisation viability score should not be interpreted as a measure of whether a destination will actually be commercially viable (it may well be, as a result of car travel), but rather as a measure of how well-adapted the surrounding area is to supply it with sufficient utilisation on the basis of active transport.

6. Conclusions

In this paper we describe methods for modelling the addition of destinations around ACs in established areas identified as the focus for implementation of x-minute policies. This extends the existing literature, which has focused on measuring current levels of accessibility, to enable the modelling of scenarios involving new destination locations. Our methods and findings are relevant in many parts of the world where

the 'x-minute city' is under consideration in low-density contexts.

Applying our methods to a low-density city, we found that while adding destinations in and around defined ACs would increase walking accessibility levels across the whole of the city, the highest improvements would accrue to the parts of the city where identified ACs are already more closely clustered. However, greater improvements would accrue to areas without such clustering if safe cycling infrastructure were available to a level where it was reasonable to treat the x-minute neighbourhood as encompassing areas within 10 minutes' cycling distance from the AC, rather than solely walking distance.

However, we also found that the feasibility of increasing access to many destinations to meet defined accessibility targets was jeopardised because there would be insufficient people in their walking catchment to make those destinations viable. The highest levels of underutilisation by residents within a walkable catchment corresponded to the lowest levels of dwelling. Again, we found that levels of underutilisation would decrease if people within an x-minute cycling distance, rather than an x-minute walking distance, were considered as the destination catchment. Our findings from the underutilisation analysis underscore the importance of combining policies aimed at increasing both density and cycling with policies for development of x-minute cities in order to maximise their success.

CRediT authorship contribution statement

Steve Pemberton: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tayebeh Saghapour:** Writing – review & editing, Writing – original draft, Investigation. **Billie Giles-Corti:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Mahsa Abdollahyar:** Writing – review & editing, Methodology, Investigation. **Alan Both:** Writing – review & editing, Validation, Methodology, Investigation. **Daniel Pearson:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Carl Higgs:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Afshin Jafari:** Writing – review & editing, Methodology, Investigation. **Dhirendra Singh:** Writing – review & editing, Methodology. **Lucy Gunn:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **James Woodcock:** Writing – review & editing, Funding acquisition, Conceptualization. **Belén Zapata-Diomed:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

Nothing to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cities.2025.106717>.

Data availability

Data, except where non-public and not able to be made available, is in our data repository at <https://osf.io/54zpc>. See also our code repository at <https://github.com/jibeproject/20mnMelbourne>.

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